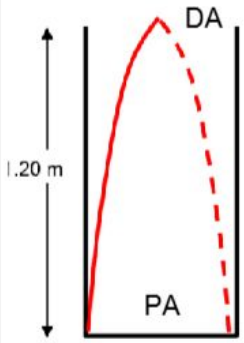
	$= 820\ J$
4	
(a)	Coherent sources / constant phase relationship / same frequency
	Same amplitude / amplitude very close to each other
(b)(i)	$\frac{\lambda}{a} = \frac{\Delta x}{D}$ (Δx is fringe separation)

	$a = \frac{633 \times 10^{-9} \times 2.50}{(1.85 \times 10^{-2})/3}$
	$= 2.566 \times 10^{-4}$ m = 0.257 mm (3 sf)
(b)(ii)	*a* increases, but λ and *D* remains the same.
	As ($\Delta x = \frac{\lambda D}{a}$), Δx decreases, hence the distance between the fringes decreases.
(b)(iii)	Hence, the fluctuation in *n* will result in random variations / changes / fuctuations in the wavelengths from each slit. *n* increases, λ decreases; *n* decreases, λ increases.
	The interference pattern will shift laterally with time / describe how positions of maxima will change / describe how distances between maxima changes OR Contrast of the interference pattern will decrease.
(c)(i)	
1.	PA and DA labelled correctly
2.	Pressure variation graph correct. (Need not touch the ends of pipe)
(c)(ii)	$L = \frac{\lambda}{4}$ $f_o = \frac{v}{4L}$ $= \frac{348}{4 \times 1.20}$ $= 72.5$ Hz
(c)(iii)	$L = \frac{3\lambda}{4}$ $f = \frac{3v}{4L} = 3f_o$ $= \frac{3 \times 348}{4 \times 1.20}$
	$= 217.5$ Hz $= 218$ Hz